

Data and Modeling Proposal

1. Group Members: Jackson Pond, Isaac Smith, Henry Tullis, Wyatt Wimmer

2. What will the project be?

We will be analyzing time-series transcriptomic (microarray) data from yeast cells undergoing oscillatory metabolic cycles (“respiratory bursts”).

3. Questions you will answer

We will utilize dimensionality reduction and clustering methods to visualize and understand our dataset and the temporal profiles identified to address the question, “What are essential features of our dataset?”. We will use a train-test split on our dataset to fit, forecast and evaluate the models on multiple loss metrics, answering “Which time series models and parameters perform most effectively on our dataset?”. We will use inspection of trained models such as HMM states to answer, “What do our analyses reveal in conversation with the conclusions of the authors?”

4. Data we will use to answer our questions

The data consist of 36 time steps, each with log intensity values for the same set of over 9,000 transcripts, including over 6,400 yeast gene transcripts. The data were collected via microarray. The dataset was collected and analyzed by (Tu BP et al., 2005) in *Logic of the Yeast Metabolic Cycle*, available via the Gene Expression Omnibus online via the accession code GSE3431.

5. Techniques we think we will use

We will first use domain-specific preprocessing techniques used by the authors, including normalization techniques to preprocess the data and account for experimental variability. We will perform data visualization coupled with dimensionality reduction methods such as PCA or UMAP. We will apply time-series methods including ARMA, HMM, and possibly Kalman filter that learn model parameters from data. We will use hyperparameter optimization methods like grid search in conjunction with these learnable models. We may perform manual inspection of trained models such as analysis of HMM states, ARMA cycles, or learned KF dynamics.

6. Evaluation Metrics

Train-test split + likelihood-based metrics, including log-likelihood and BIC, for time-series model optimization. Train-test split + likelihood/loss based metrics such as log likelihood and distance-based loss (L1, L2) for time series forecasting. Loss, visual inspection, and comparison with creators’ conclusions + domain knowledge for non-forecasting analyses.

7. How we will divide work

Domain-specific data tasks like preprocessing may be performed by a subset of group members, with remaining members focusing on other data handling tasks like scraping and dimensionality reduction. We will each write some code, perform some analyses, and engage in writing.